

AI-BASED TOMATO SORTING AND SHELF-LIFE ESTIMATION SYSTEM

AI-Based Tomato Ripening Stage Detection and Sorting System with Shelf-Life Estimation for the Commonly Consumed Padma Variety in Sri Lanka Using Computer Vision and Conveyor Automation

IIT Group No 16

UWU/IIT/21/017 | UWU/IIT/21/019 | UWU/IIT/21/062

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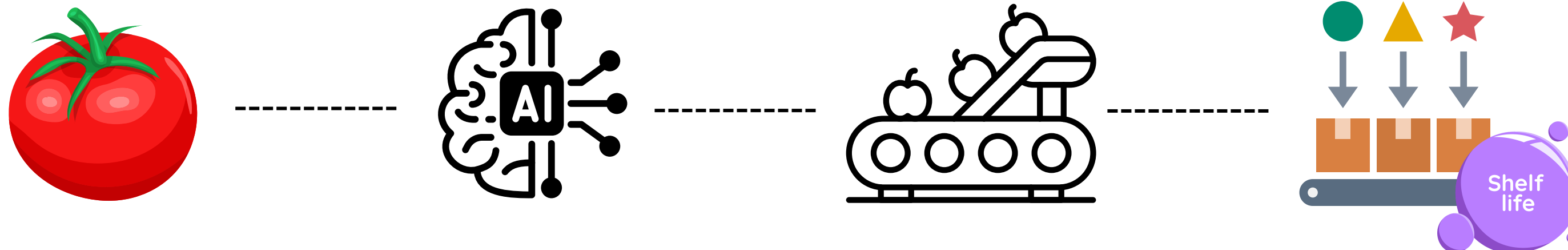


INTRODUCTION

Project Description

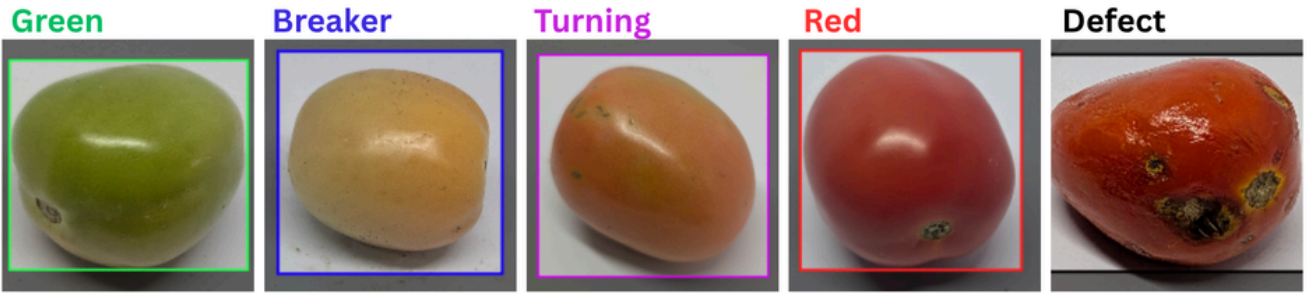
This research presents the efficiency and consistency of Padma tomato grading by introducing an AI-integrated automation system that predicts shelf life, reduces post harvest losses and labor dependency, and boosts profitability for Sri Lankan farmers.

- AI and vision with conveyor for ripeness, defects, and shelf life.
- Automated sorting via a servo actuated mechanism.
- Deep model identifies ripeness, defects, and shelf life real time.
- KPI dashboard for accuracy, throughput, and defect rate monitoring.



INTRODUCTION

Project Description



Stage 1 (Green)	Uniform green colour, firm texture	100% Green
Stage 2 (Breaker)	Green-yellow transition, <30% red area	Green - Yellow
Stage 3 (Turning)	Orange to pink surface, 30–80% red coverage	Partial Red/Orange
Stage 4 (Red Ripe)	More than 80% red surface, softer texture	Deep Red

Defects (Cracks, bruising, rot) will be diverted off the conveyor for immediate processing or rejection

INTRODUCTION

Background Study



Economic Importance



Manual Grading Challenges



Inconsistent Quality and Shelf Life



Economic Importance

Tomatoes are a high-value cash crop essential for local consumption and potential export markets.



Manual Grading Challenges

The current sorting process is labor-intensive, slow, and highly susceptible to human error, leading to inconsistent product quality.

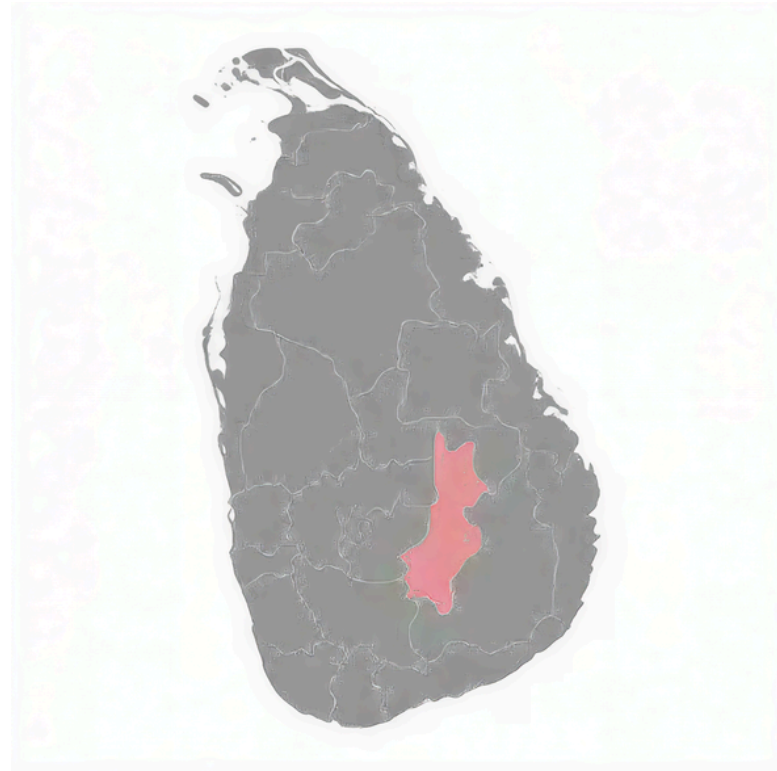


Inconsistent Quality and Shelf Life

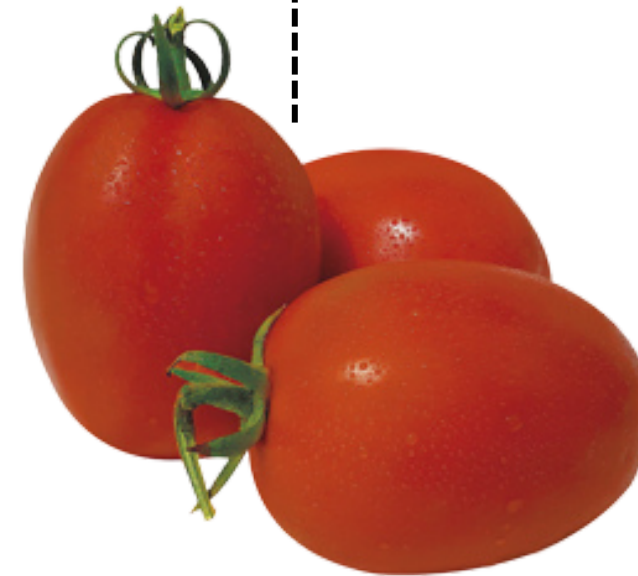
Subjective grading leads to mixed batches, lower quality, and unpredictable shelf life with higher financial risk.

INTRODUCTION

Why “Padma” ?



- Badulla
- Bandarawela
- Welimada
- Mahiyangana

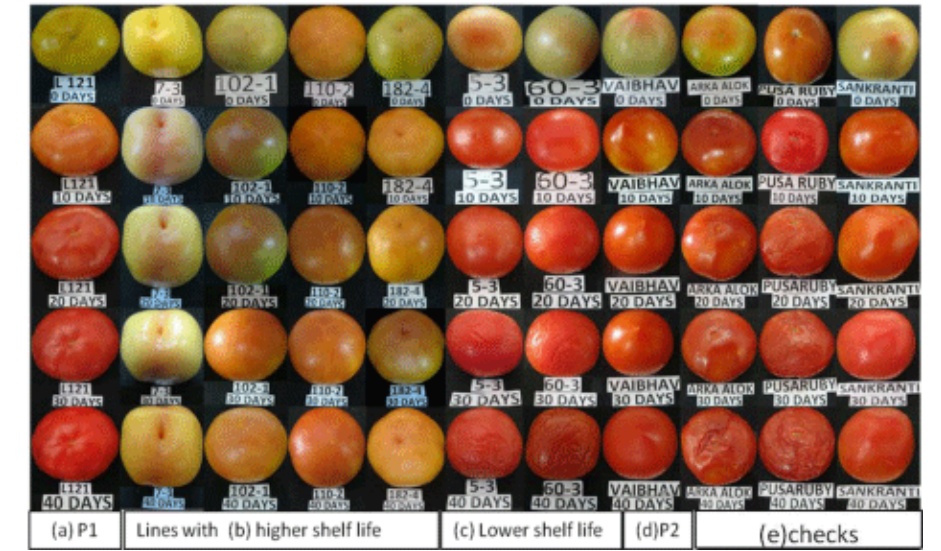


- T-245
- Tomato-1014
- Red-Bullet
- King-Kong,
- **Padma**
- Platinum

- Dominant Local Cultivar: Key crop in Badulla & Nuwara Eliya, preferred by farmers for its high yield and bacterial wilt resistance.
- High Post-Harvest Losses: Perishable climacteric fruit susceptible to microscopic skin damage and rapid decay from manual sorting.
- Established Research Baseline: Post-harvest ripening stages and 31-day shelf life benchmarked under local conditions by Abekoon et al. (2024).

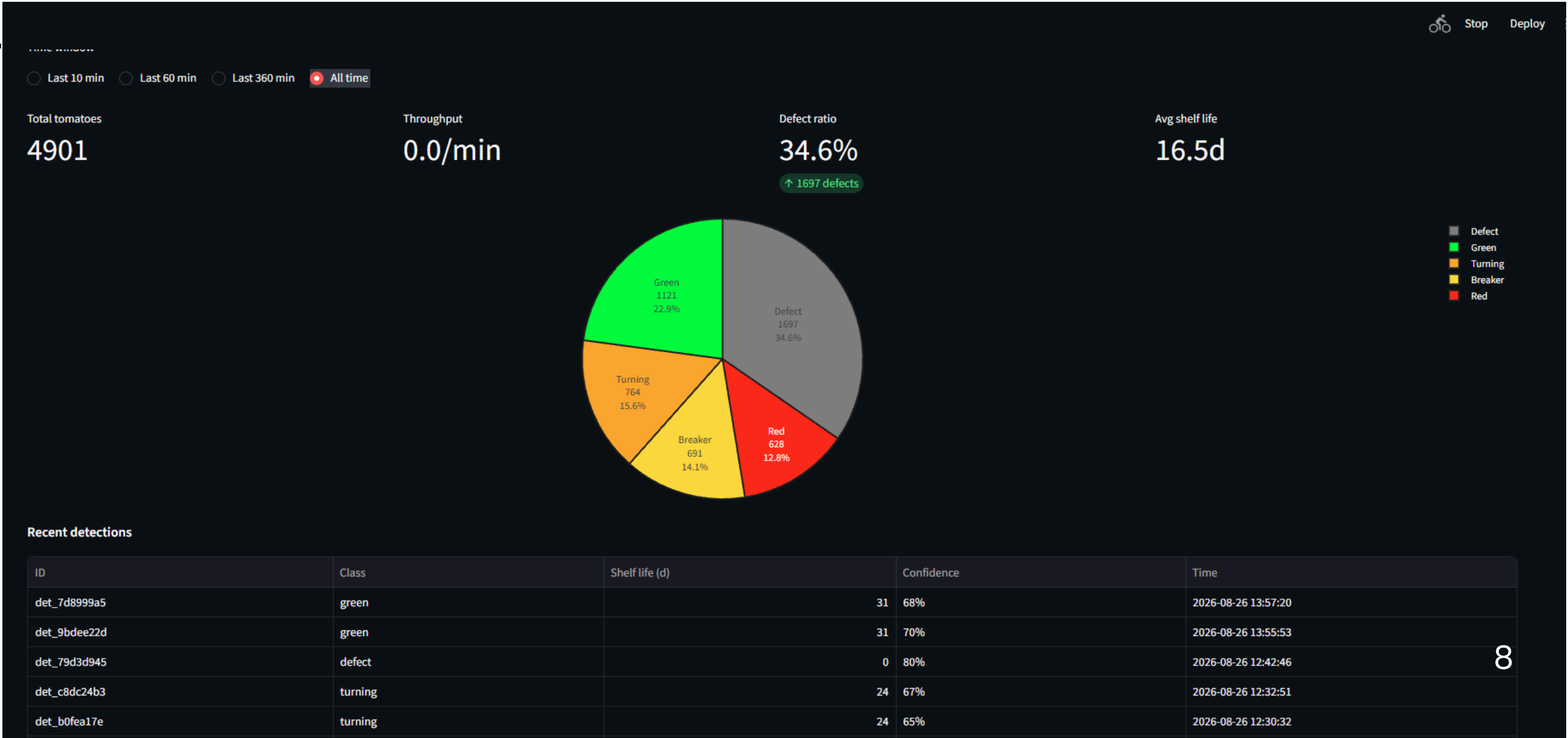
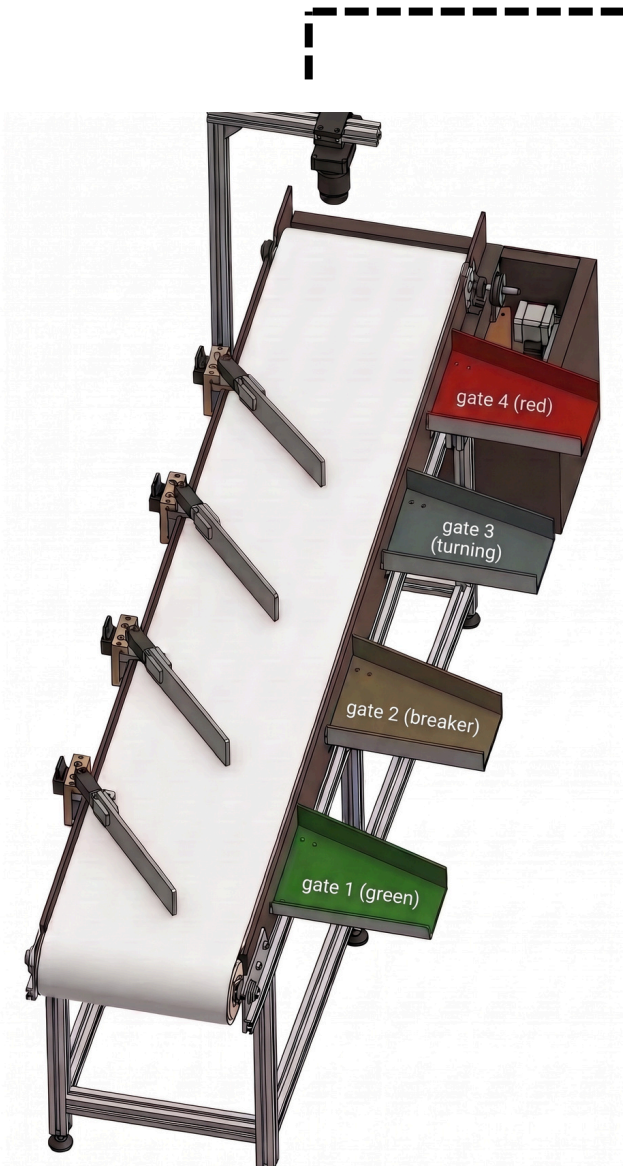
RESEARCH PROBLEM

- 1 Manual tomato sorting is labor intensive, time consuming, and prone to subjective judgments, leading to mixed quality and various shelf life in market batches.
- 2 Post-harvest losses can reach up to 20% due to poor or inconsistent grading (FAO, 2022).
- 3 There is a lack of automated solutions optimized for Padma tomatoes, especially for detecting multiple ripening stages and defects under real-time, local conditions.
- 4 Addressing these gaps is essential for enhanced productivity, reduced waste, and improved profitability for Sri Lankan farmers.



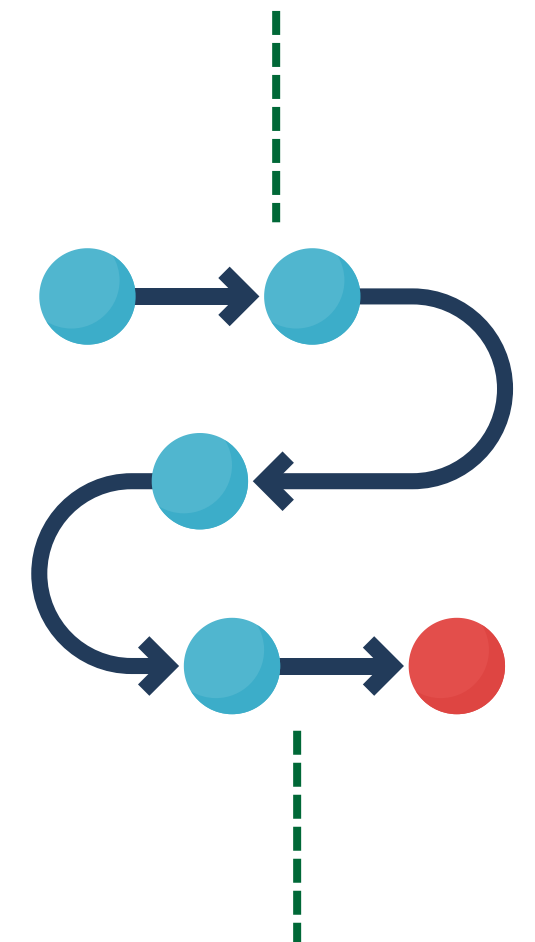
FINAL GOAL

To design and develop an AI-based real-time system to detect, classify, and sort Padma tomatoes based on ripeness stage and visible defects, and to estimate remaining shelf life.



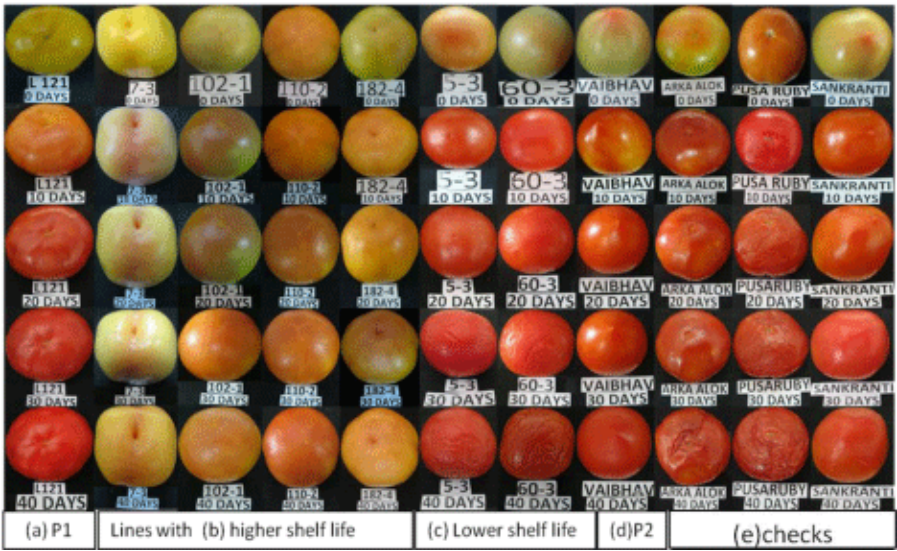
OBJECTIVES

- To collect and label a local dataset of more than 9000 Padma tomato images covering four ripeness stages and one defect class under real farm and post-harvest lighting conditions.
- To develop and train a Deep Learning model that performs multi-class classification and defect detection.
- To design and construct a mechanical conveyor prototype mechanism that automatically sorts tomatoes according to detected ripeness stage and defects.
- To integrate and evaluate a shelf-life prediction and KPI monitoring system that estimates remaining freshness (in days) and displays class distribution, defect ratio, and throughput on a live dashboard.



MOTIVATION

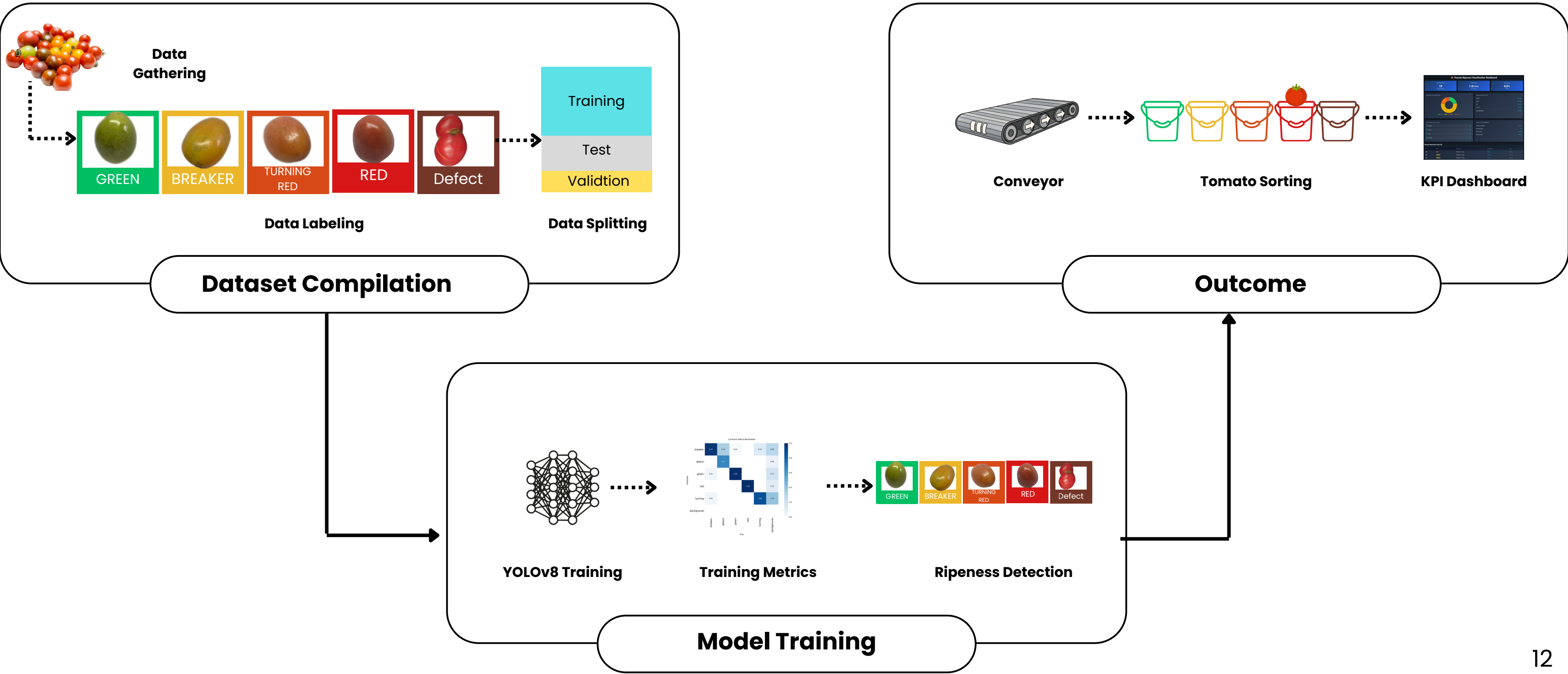
- High Post-Harvest Losses in Local Produce
- Limitations of Manual Visual Grading
- Gap in Cultivar-Specific AI & Conveyor Automation
- Need for an Integrated System



LITERATURE REVIEW

Study / Source	Methodology & Scope	Key Limitation / Gap	Relevance to Our Work
Abekoon et al. (2024)	Image processing on Sri Lankan Padma variety (7 stages)	Offline classification of static images; no AI object detector or conveyor sorting	Groundwork for Padma ripening stage duration & shelf-life mapping
Moya et al. (2025)	YOLOv8 model deployed on conveyor sorting line	Limited to 3 broad classes; no shelf-life prediction or KPI tracking	Baseline proof for real-time single-stage conveyor detection feasibility
Geetha et al. (2025)	Deep learning for ripeness + shelf-life estimation	Generic datasets; error-prone constant-speed actuation	Justifies combining vision-based sorting with live shelf-life metrics

METHODOLOGY OVERVIEW



METHODOLOGY

DATA ACQUISITION & PREPARATION

**CLD Farm Fresh
(Bandarawela)**

**Controlled white-
box + LED setup**

**>9,600 images, 5
classes**

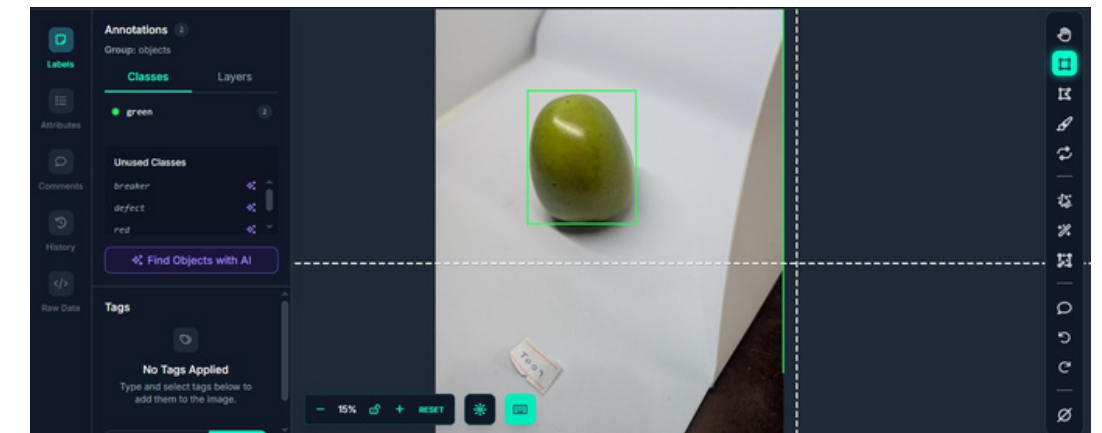
**Roboflow
annotation**

MODEL DEVELOPMENT & TRAINING

YOLOv8m

**Single-model
approach**

**PyTorch/Ultralytics
training**



Classes & Tags

Classes 5 Tags 0			What is a class?	Lock Classes	+ Add	Modify Classes
Search classes...			Sort By: Class Ascending			
COLOR	CLASS NAME	COUNT				
	breaker	2,011				
	defect	2,025				
	green	2,002				
	red	2,020				
	turning	2,070				

METHODOLOGY

HARDWARE INTEGRATION

Conveyor system



**ESP32
Microcontroller**



Actuator & sensors

SOFTWARE DESIGN

**Real-time inference
pipeline**



**KPI dashboard
(Python Streamlit)**



**Shelf-life
estimation**

TESTING & EVALUATION

**4-phase testing
strategy**



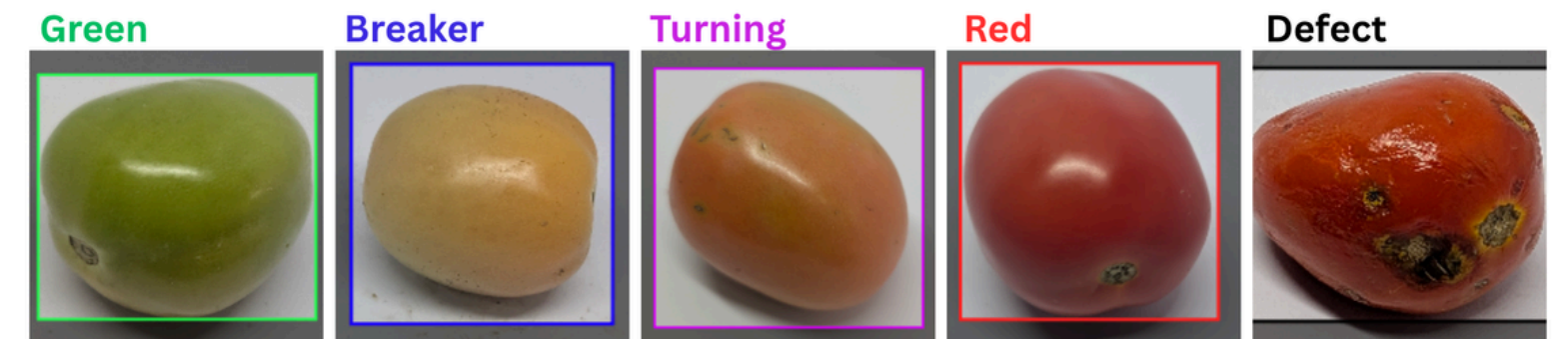
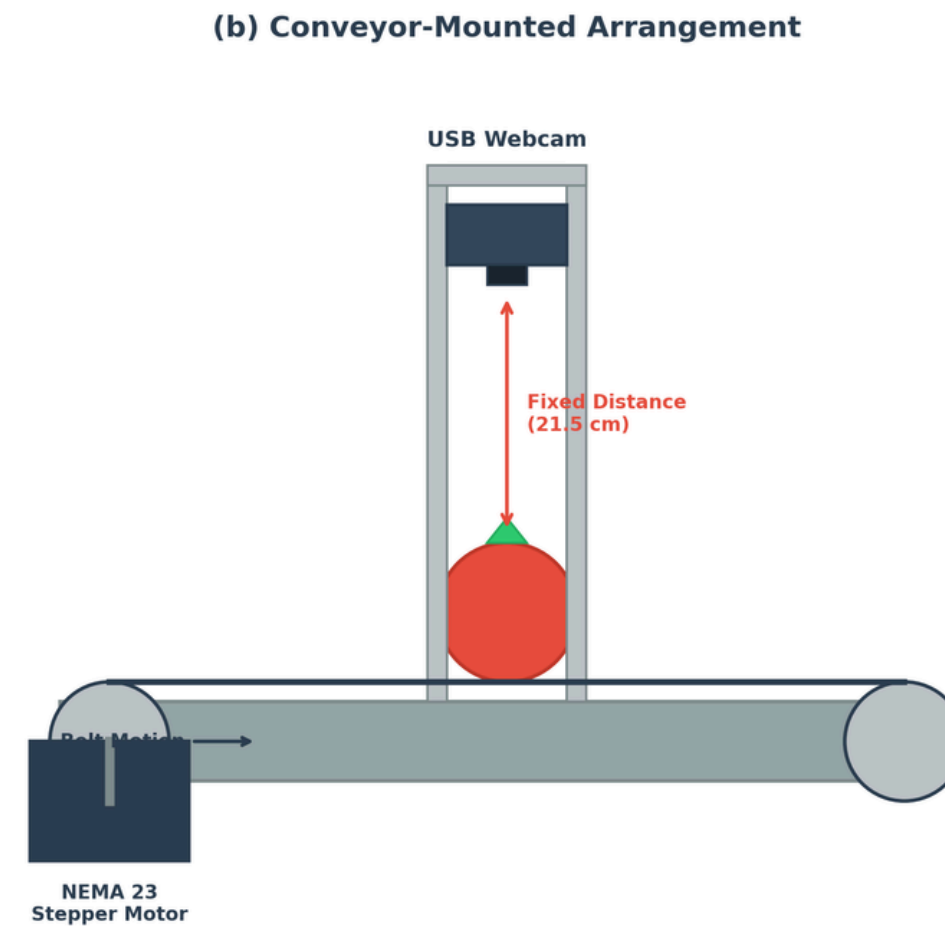
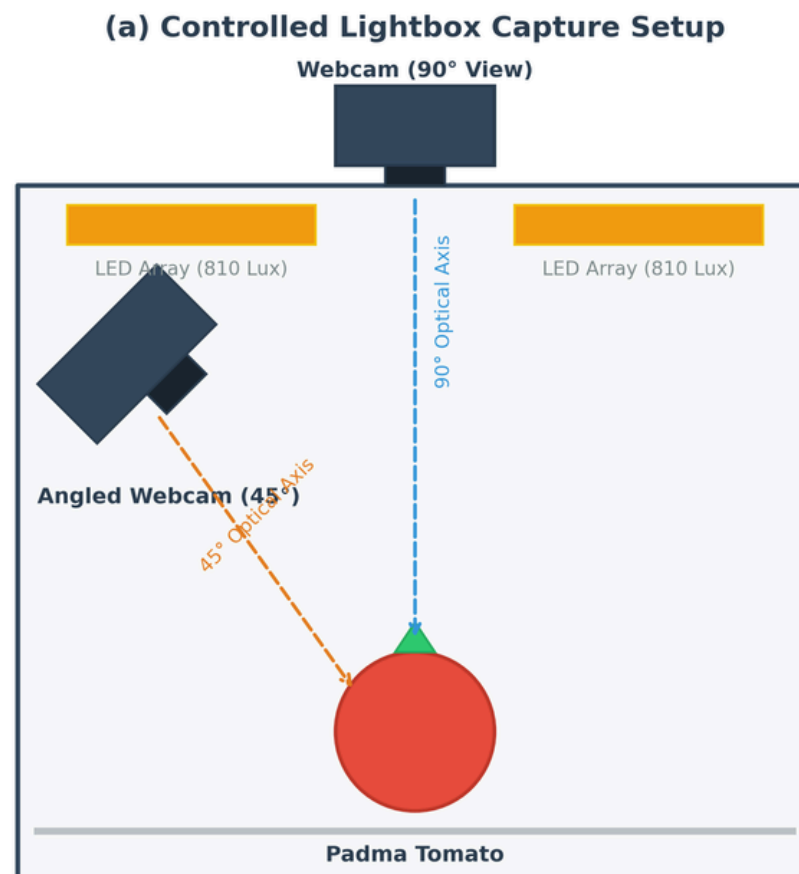
**Performance metrics
(mAP, precision)**



**Prototype Lab &
Bench Validation**

METHODOLOGY

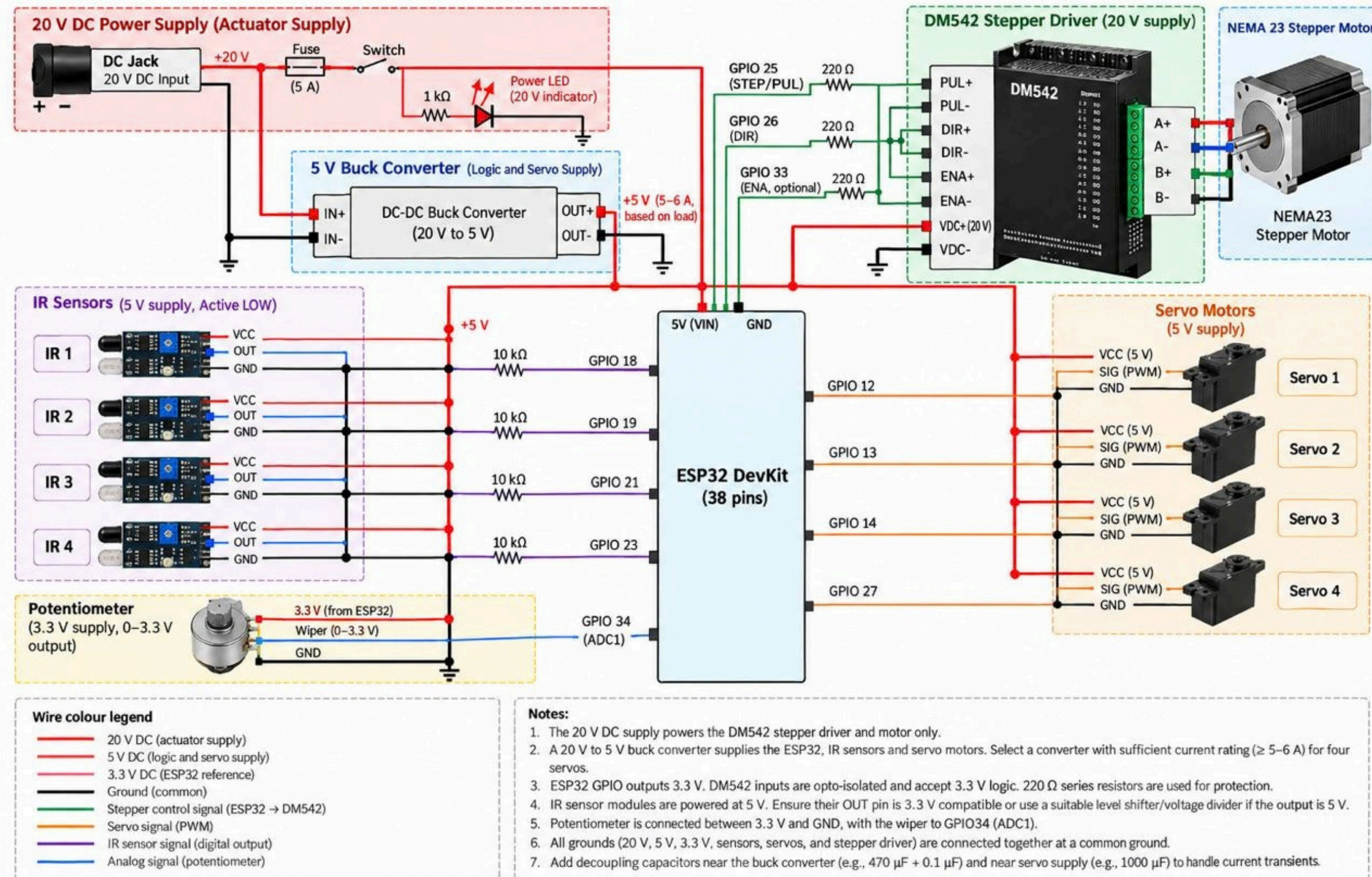
Data Acquisition & Dataset Preparation



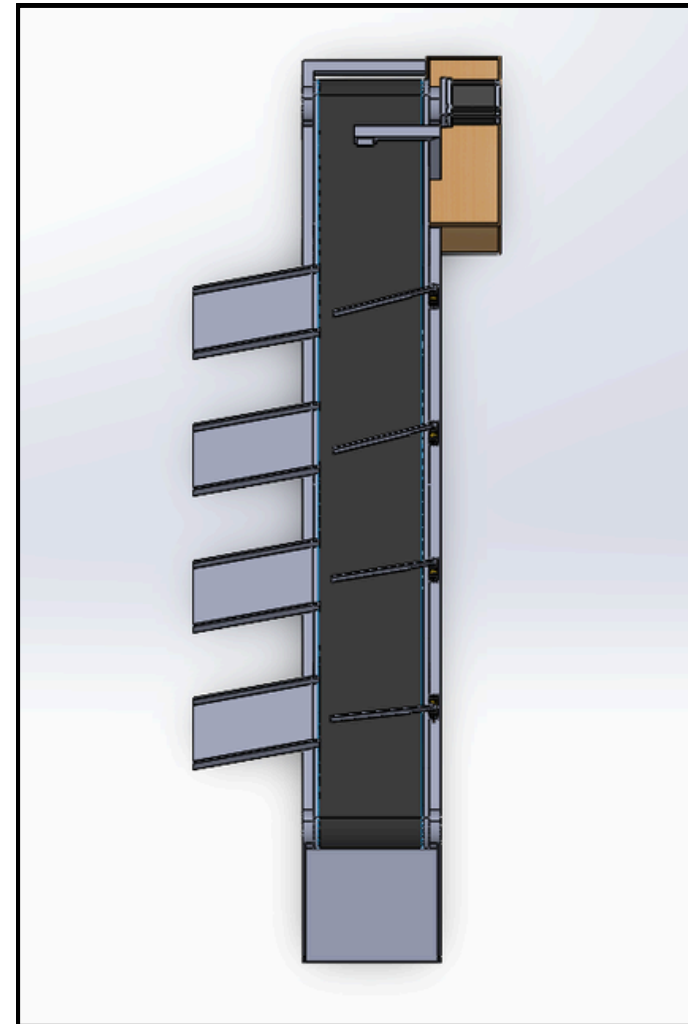
9,600+ Images, Roboflow platform, Lightbox setup, 5 Classes (Green, Breaker, Turning, Red, Defect).

METHODOLOGY

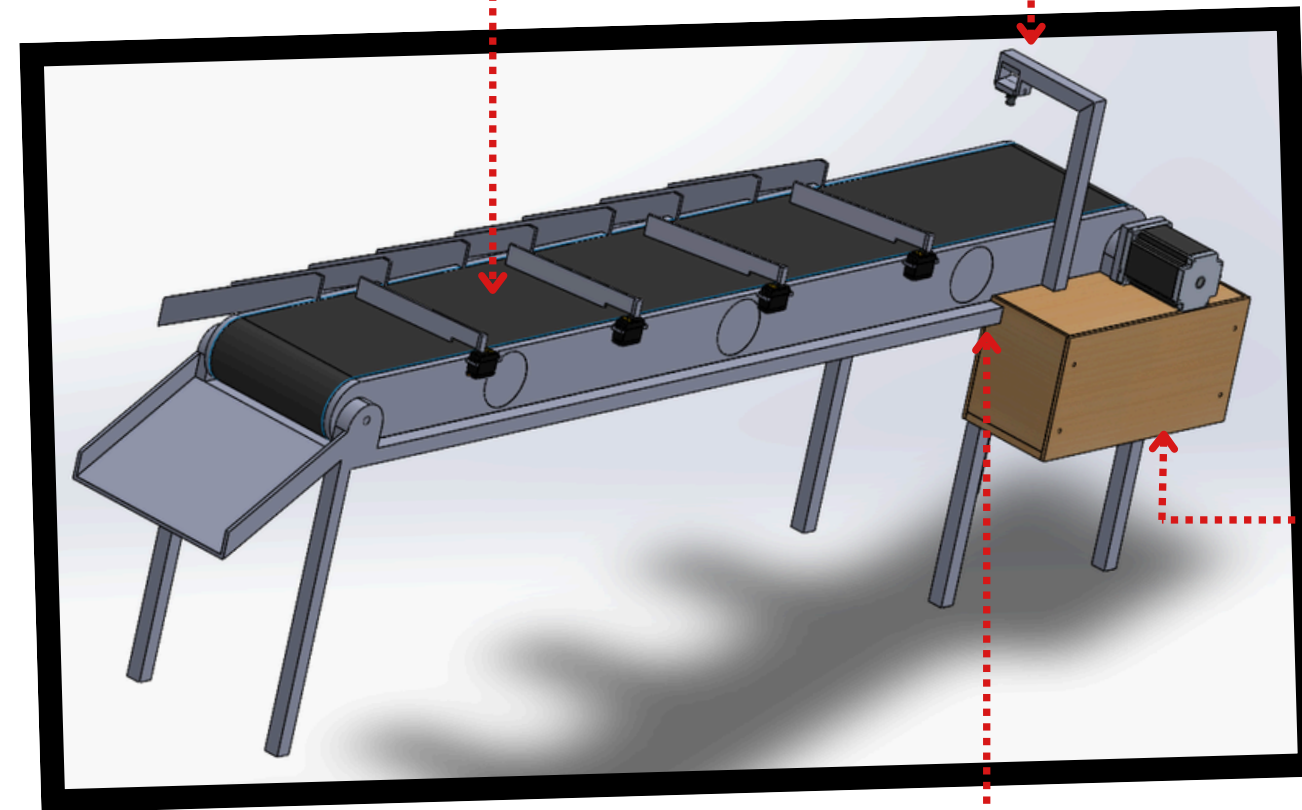
Wiring Diagram



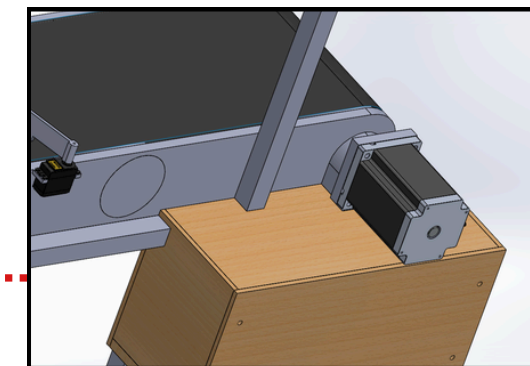
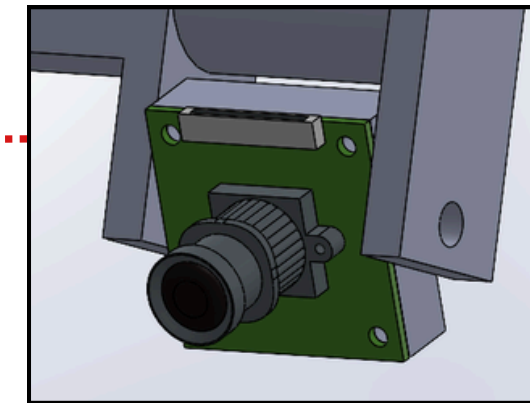
PROTOTYPE DESIGN



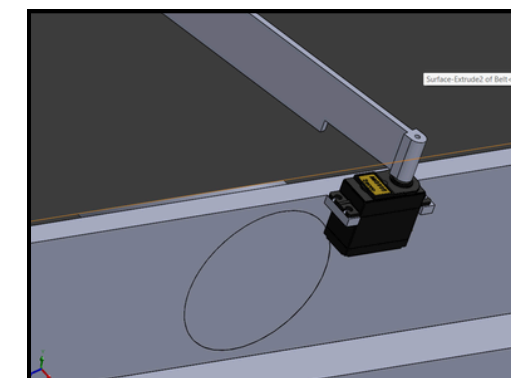
4-Stage Sorting Chutes



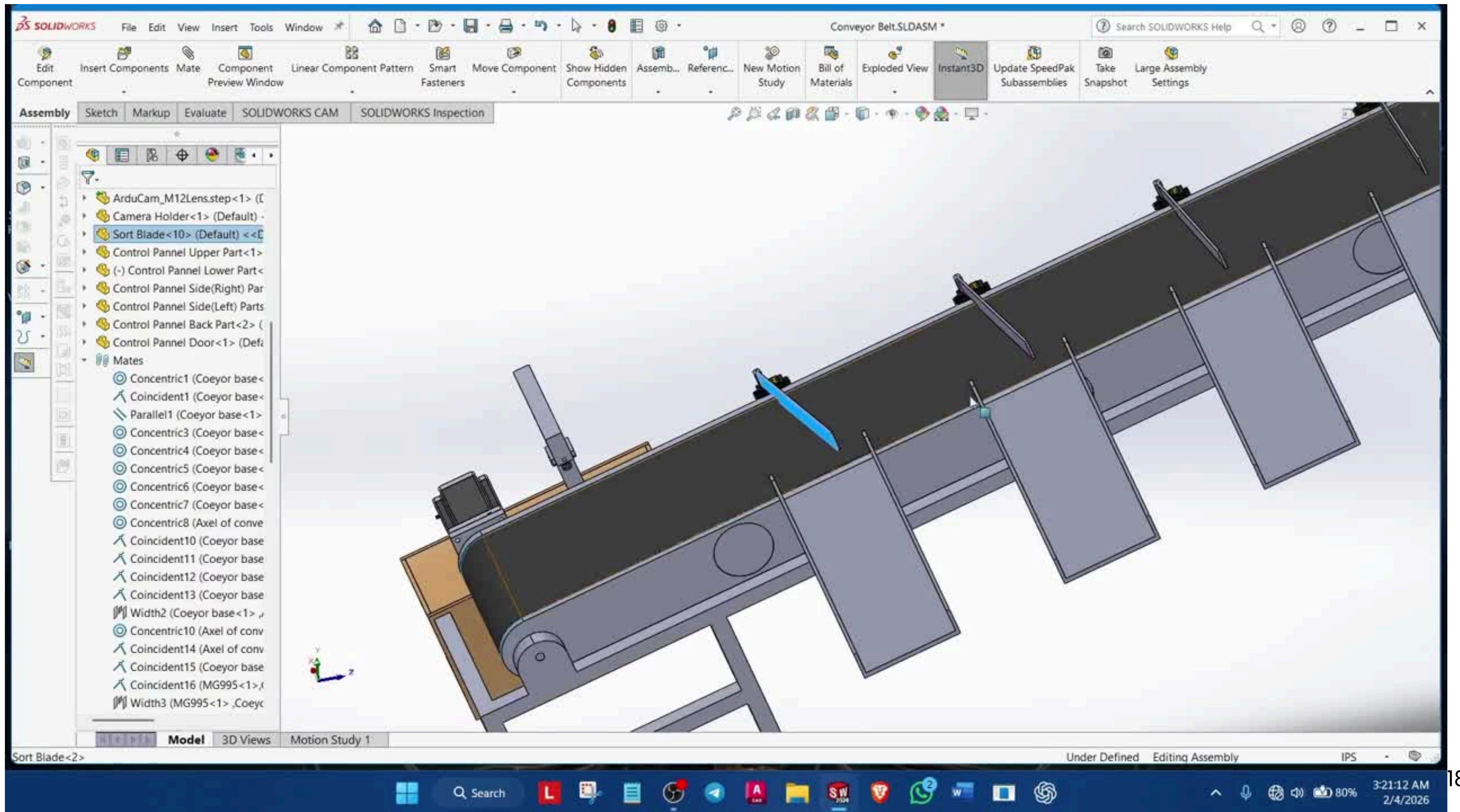
Overhead Camera Mount



NEMA 17 Stepper Drive Motor

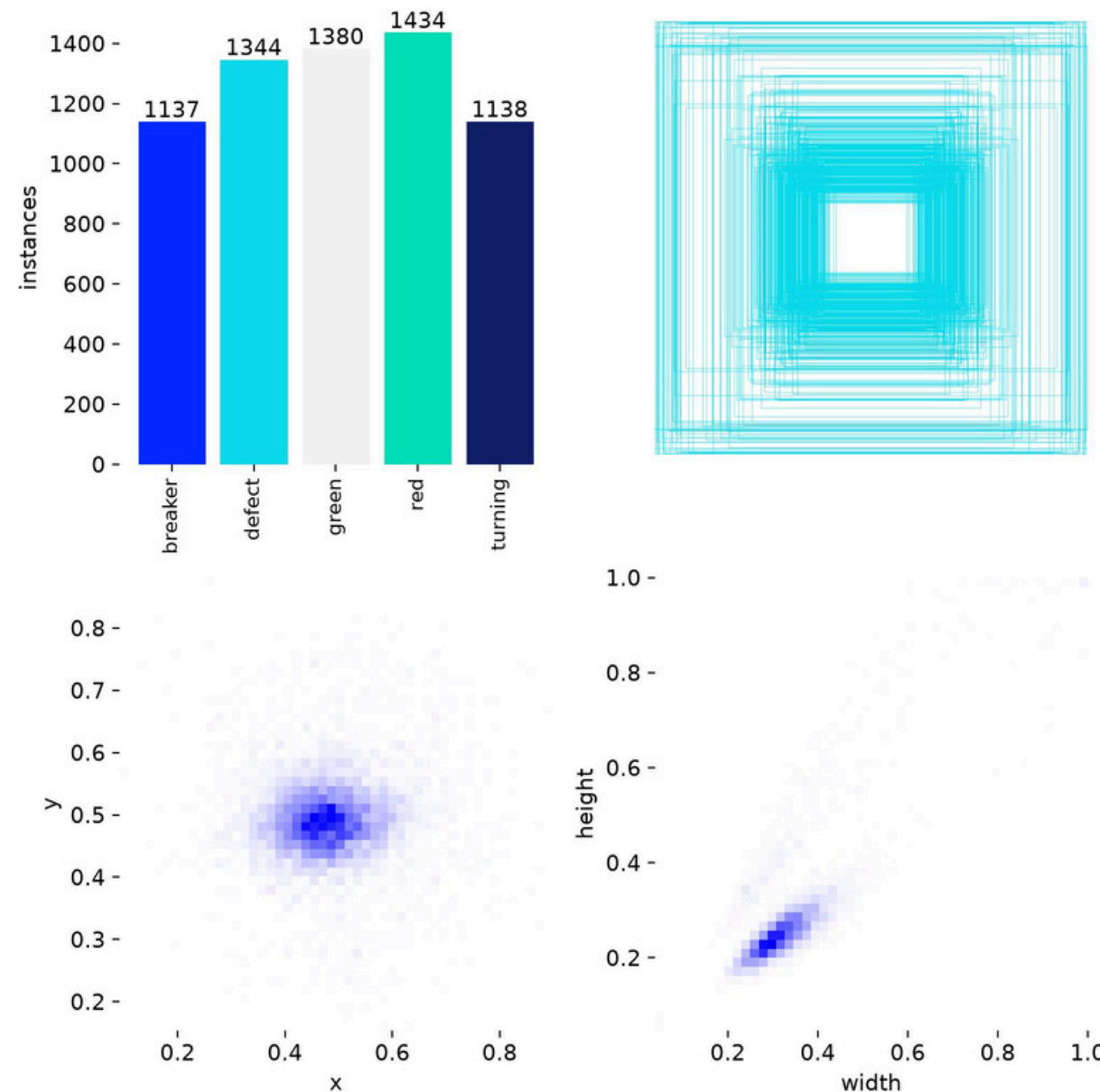


Servo Gate Actuator



RESULTS & DISCUSSION

YOLOv8 model retrained and evaluated on an expanded Padma tomato dataset (5 classes: green, breaker, turning, red, defect)

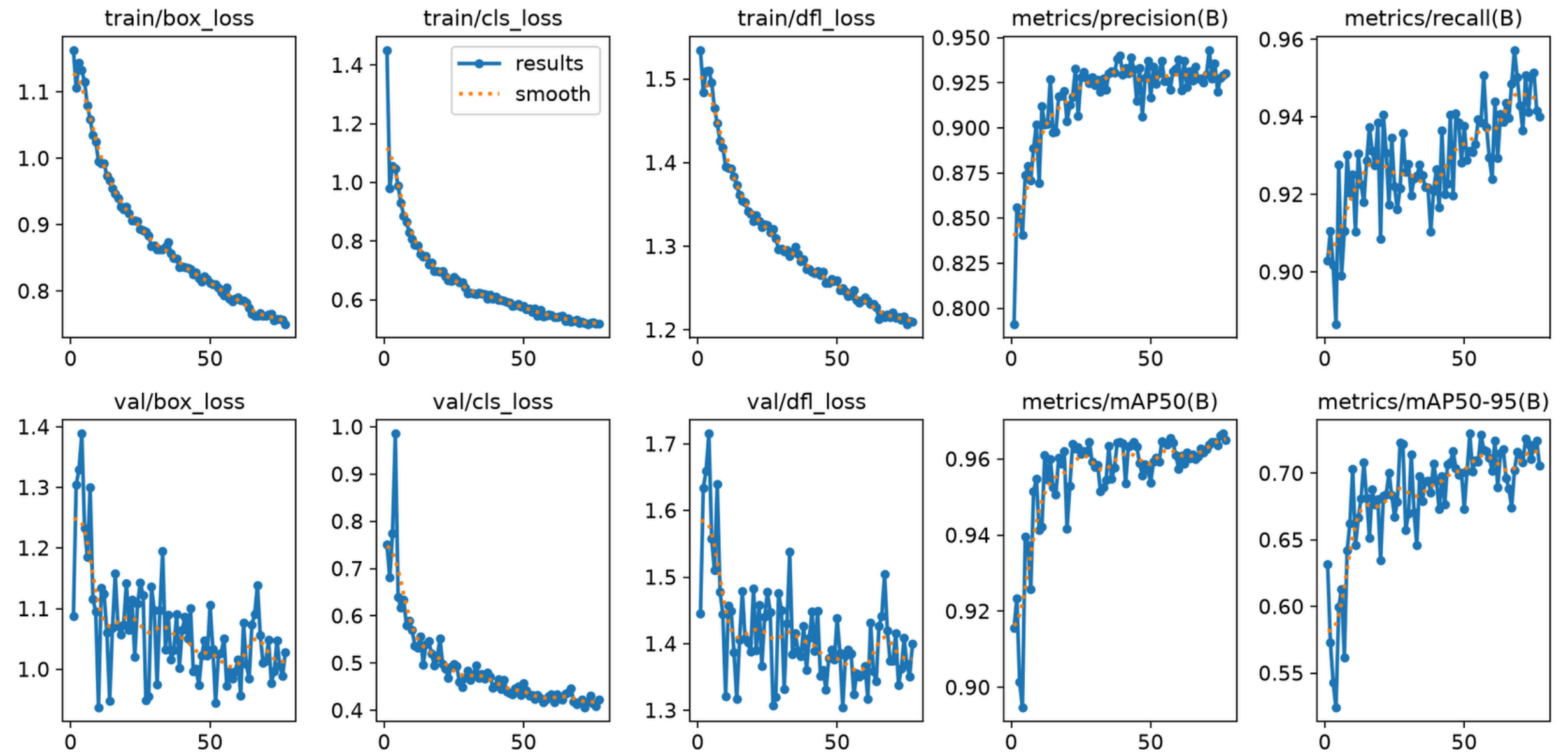


**≈11,400+ annotated
tomatoes across 5 classes**

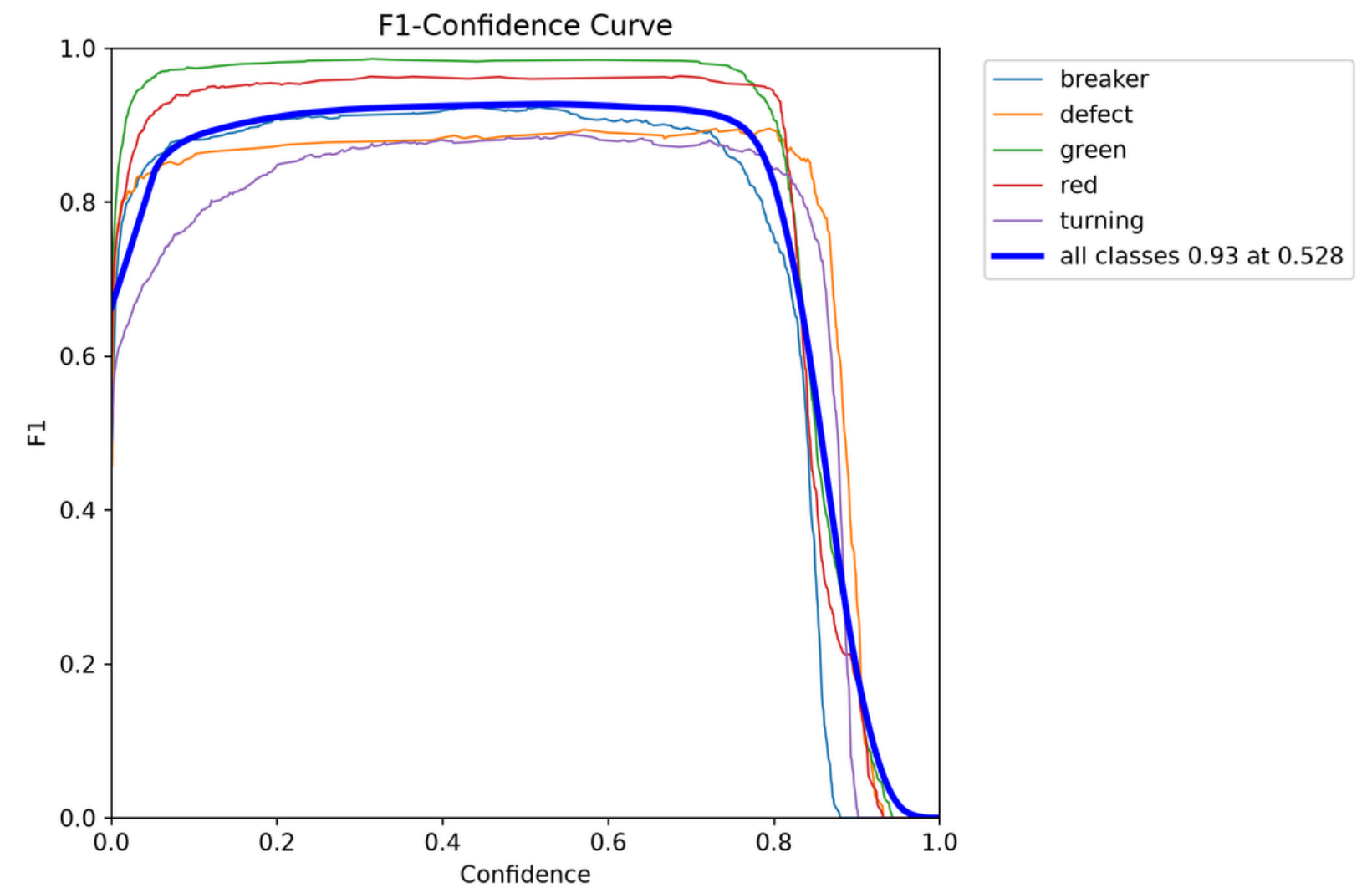
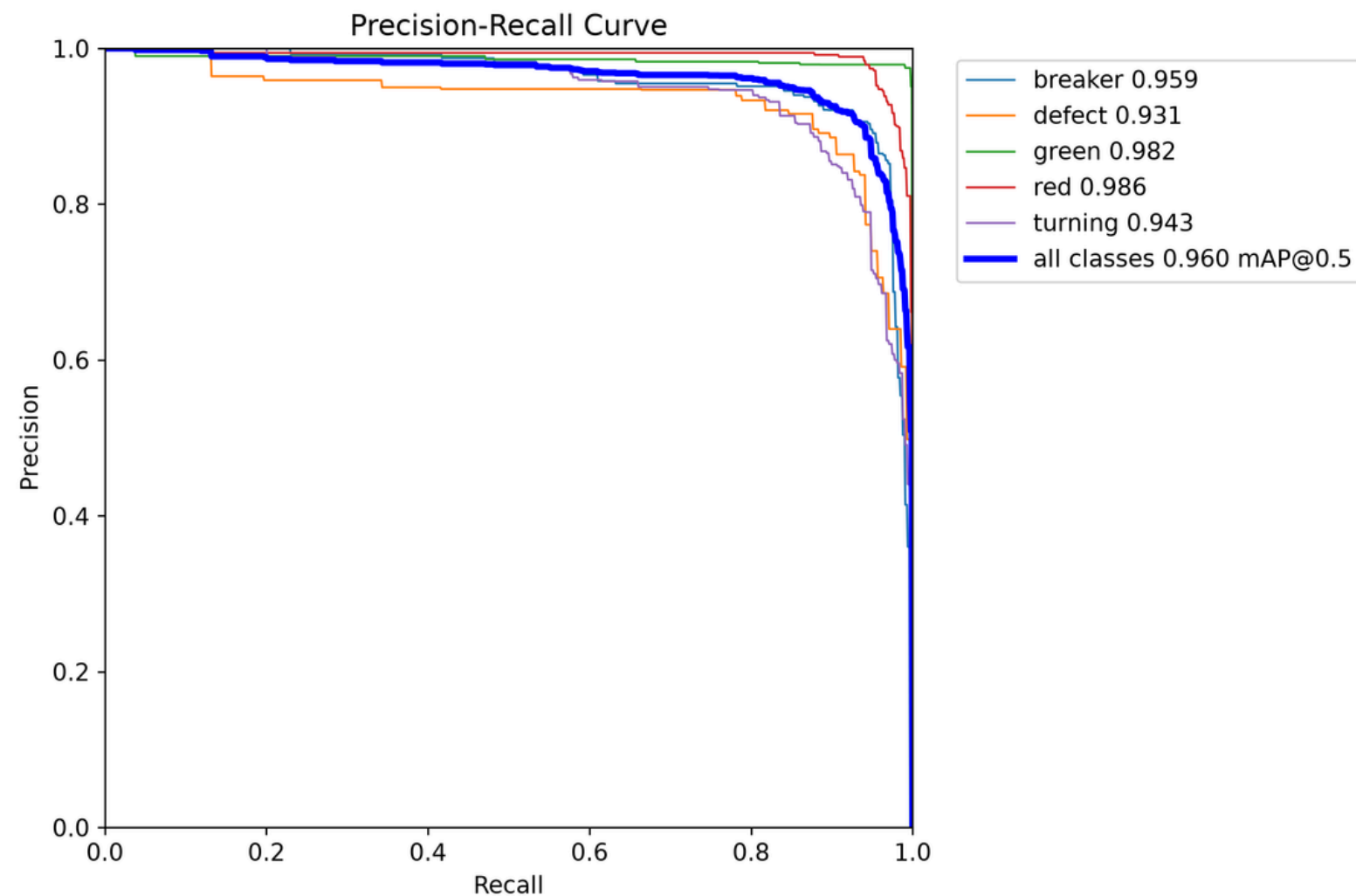
**Classes are well balanced
(~2,100–2,450 instances per class)**

RESULTS & DISCUSSION

These graphs show the model learned well on the new dataset. The training and validation losses drop quickly in the first epochs and then level off (better box placement and class predictions). Validation loss stays low and does not rise again, so there is no sign of overfitting. Precision and recall reach about 0.93–0.94, mAP50 about 0.96 and mAP50–95 about 0.70. The model makes few false detections, finds almost all tomatoes, and places the boxes accurately.



RESULTS & DISCUSSION



This Precision-Recall graph shows the model now works well for all five classes. Green and red are still the strongest (about 0.99 AP). Defect detection has improved the most, from about 0.19 to about 0.93 AP, so defects are no longer a weak point. Overall mAP@0.5 is about 0.96, and the F1 curve peaks at about 0.91 at a confidence of about 0.53.

Results Summary -Previous vs. New Model

Metric	Previous	New
Annotated tomatoes	≈9,300	≈11,400
Instances per class	~1,500–2,000	~2,100–2,450
mAP@0.5 (all classes)	0.903	0.960
Defect AP	0.185	0.93
Best F1 score	0.89	0.91
Recall (all classes)	0.95	0.98

KEY TAKEAWAYS

- Larger, more balanced dataset: about 2,000 more annotated tomatoes, and every class now has more than 2,100 instances.
- Higher overall accuracy: mAP@0.5 rose from 0.90 to 0.96.
- Defect detection greatly improved: defect AP went from 0.19 to 0.93, so it is no longer the weakest class.
- More stable and confident: the F1 score is higher and the best confidence threshold rose from 0.39 to 0.53.

CONCLUSION

The new YOLOv8 model is more accurate and reliable across all five ripeness and defect classes. The biggest gain is in defect detection.

RESULTS & DISCUSSION

- **Final Outcome -: Tomato Shelf Life Estimation**

- Experiment conducted under room temperature.(between 22°C-24°C)
- Observations recorded daily morning time over 31 days.
- Shelf life estimation for each ripening stage:

Tomato Stage	Duration of Stage (Days)	Shelf Life (Days)
 Green	2	31
 Breaker	5	29
 Turning Red	12	24
 Red	12	12
Total	31	—

- Purpose: **Helps in determining optimal sorting and storage time.**





Day 01



Day 02



Day 03



Day 04



Day 05



Day 06



Day 07



Day 08



Day 09



Day 10



Day 11



Day 12



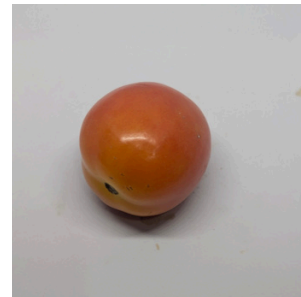
Day 13



Day 14



Day 15



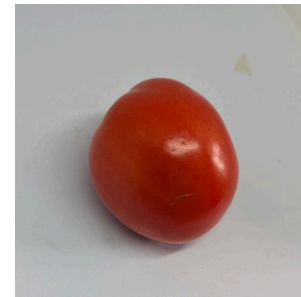
Day 16



Day 17



Day 18



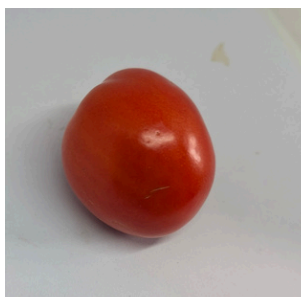
Day 19



Day 20



Day 21



Day 22



Day 23



Day 24



Day 25



Day 26



Day 27



Day 28



Day 29



Day 30

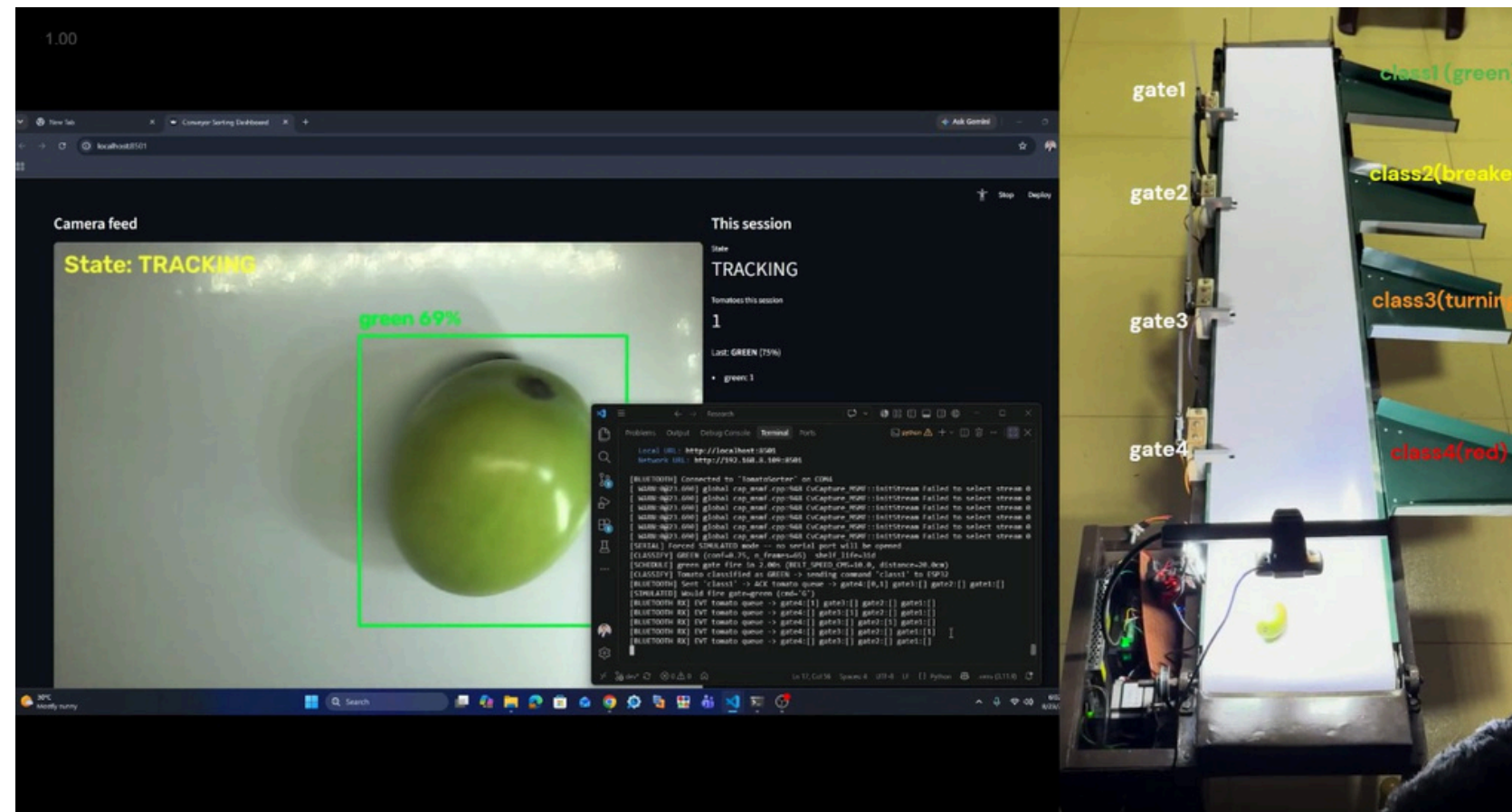


Day 31

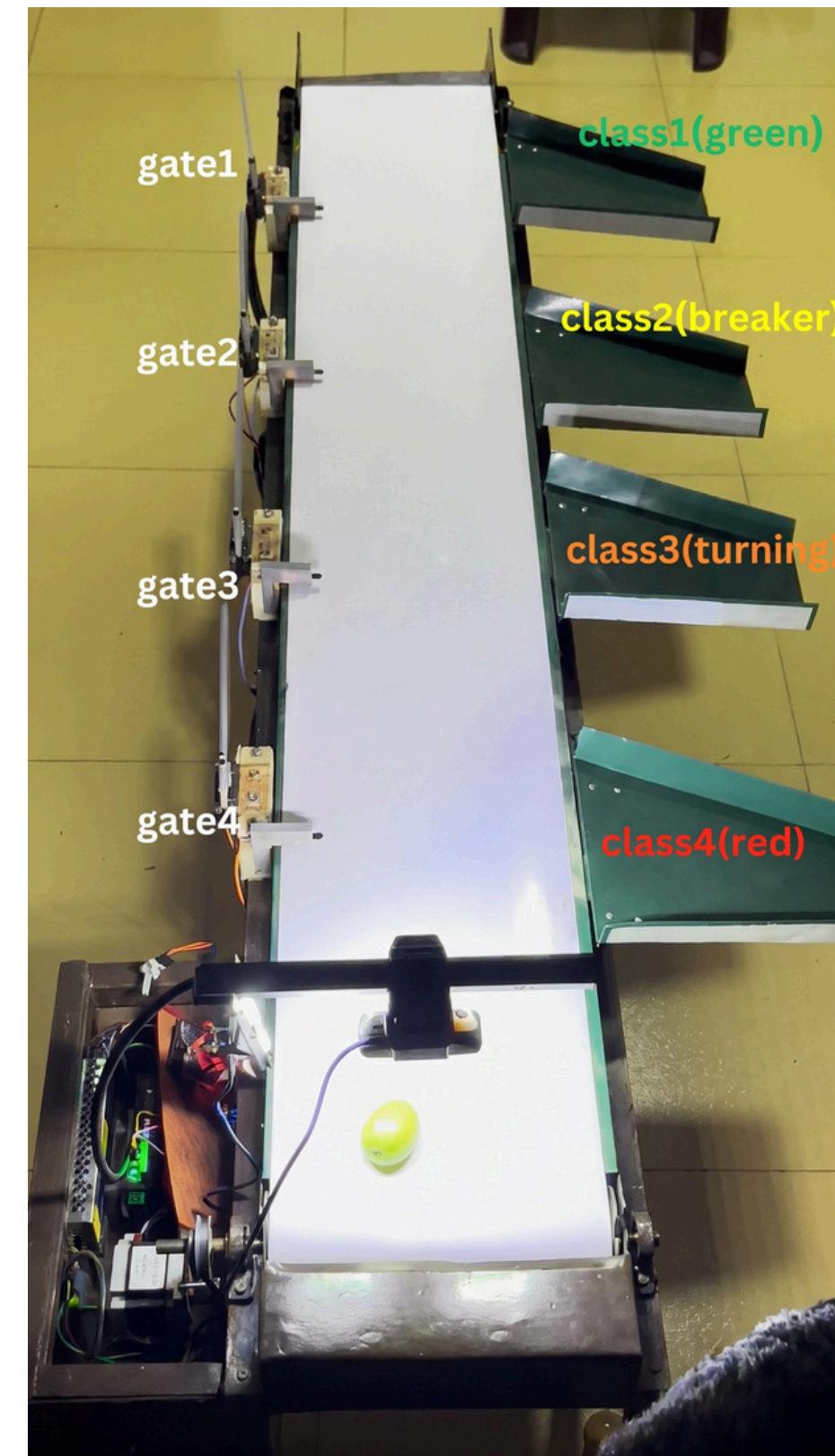
RESULTS

OUTCOME

Software Logic & Real-time Object Tracking



The system combines multi-frame OpenCV centroid tracking with speed-scaled IR sensor triggers and Bluetooth queues to ensure accurate, zero-duplicate tomato sorting without frame drops.



CONCLUSION

High-Precision Vision System

- YOLOv8m AI Model: Achieved 0.936 mAP@50 high classification accuracy.
- Ultra-Fast Speed: Operates at 6.7 ms per frame.

1

Sensor-Driven Sorting

- ESP32 Control: Event-triggered IR sensors.
- Precise Sorting: Zero fruit damage/timing errors.

2

Logistics & Shelf-Life Intelligence

- 31-Day Mapping: Tracks lifespan per stage.
- Live Dashboard: Real-time batch metrics.

3

Supply Chain Impact

- Automated Padma Tomato Grading: Replaces manual handling with low-cost AI

4

RECOMMENDATION

single level project, with no interaction using laptop,
machine can be run as a single unit without using a laptop ,
and process can done via rasbery pi or upper version , (as
a future improvement)



seperate model to remove non
tomato objects at the begening



detect the tomatos with max
stepper motor belt speed



Multi-Object Tracking & High-
Throughput Sorting



Per-Fruit Dynamic Shelf-Life
Estimation



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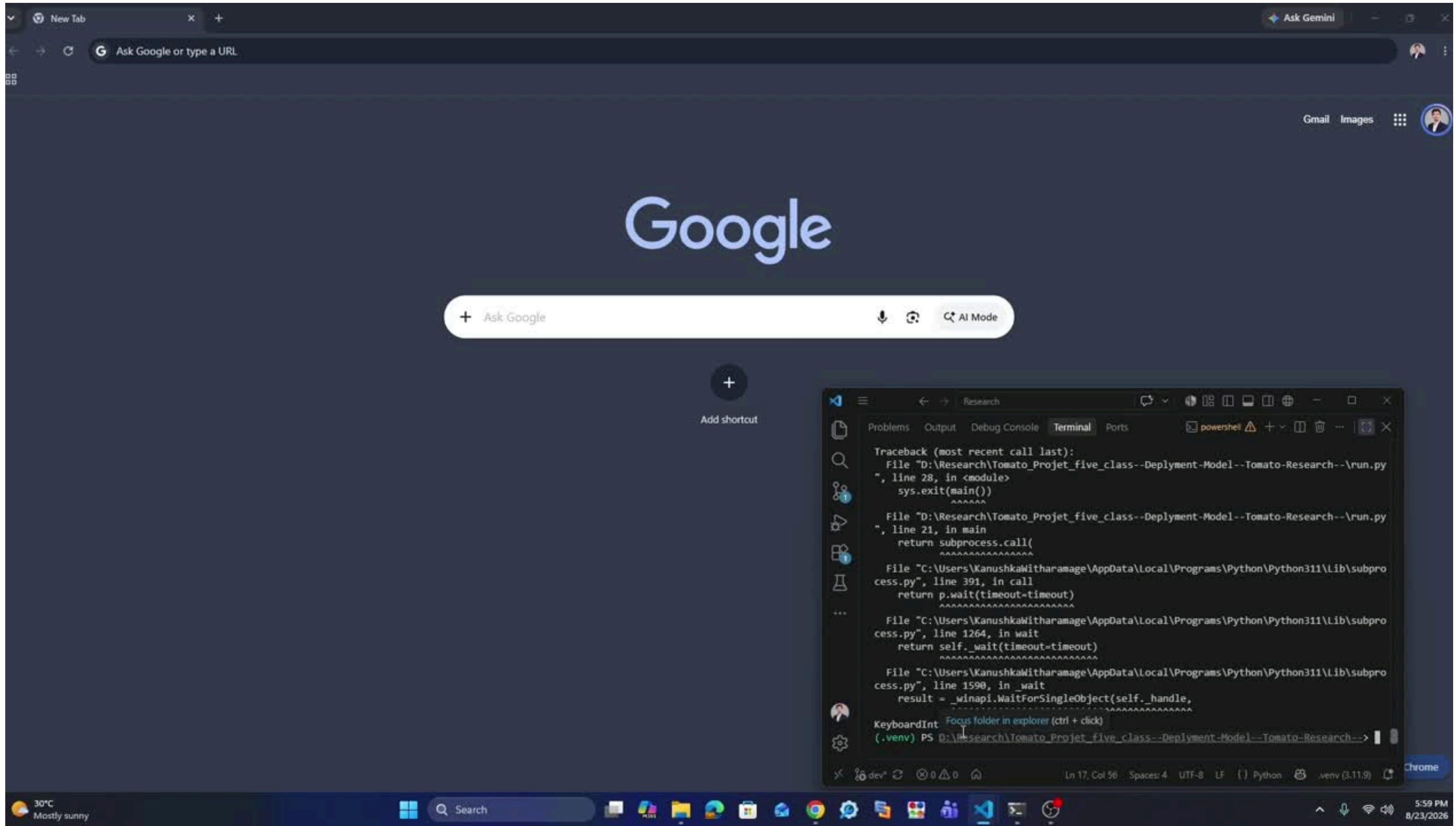
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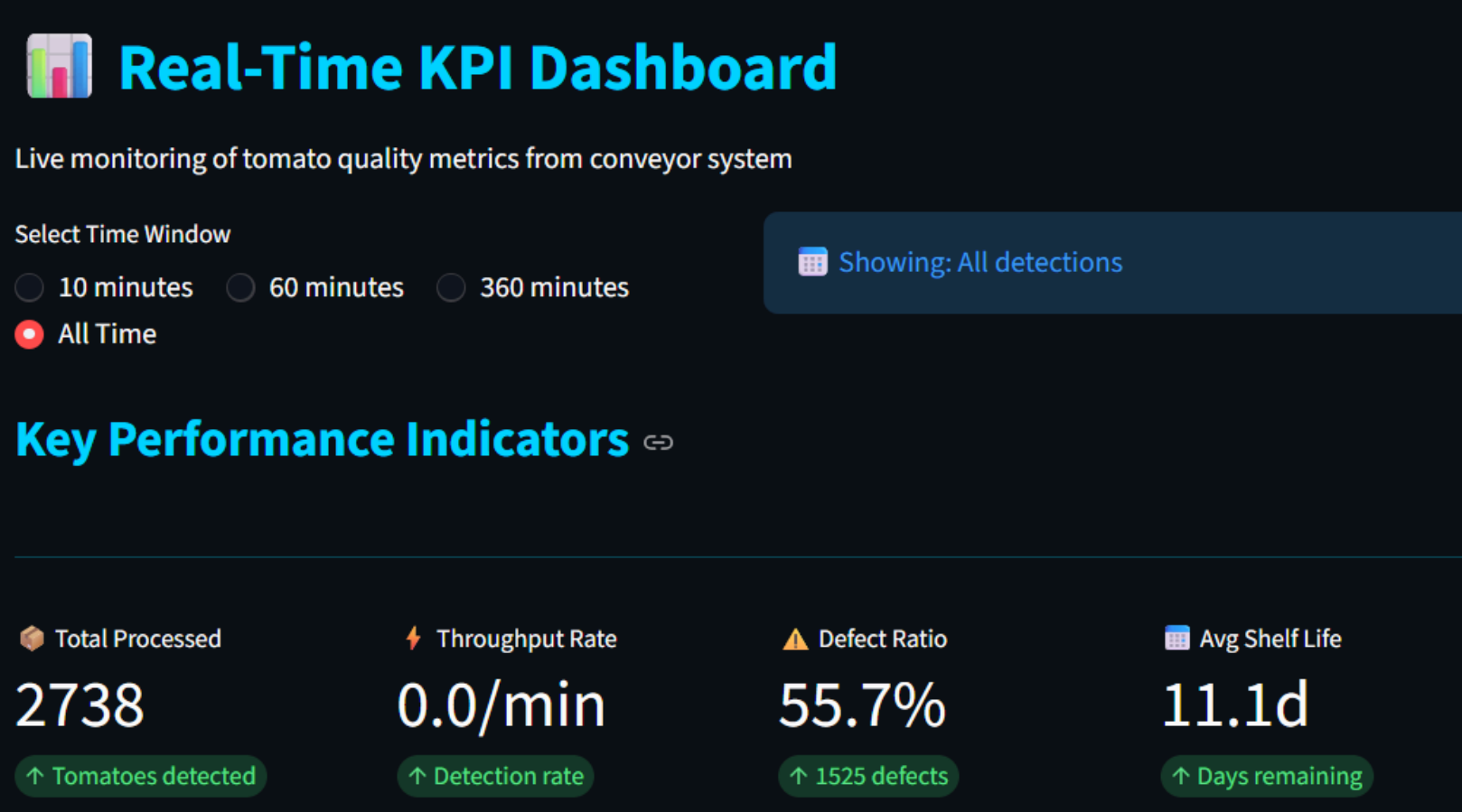
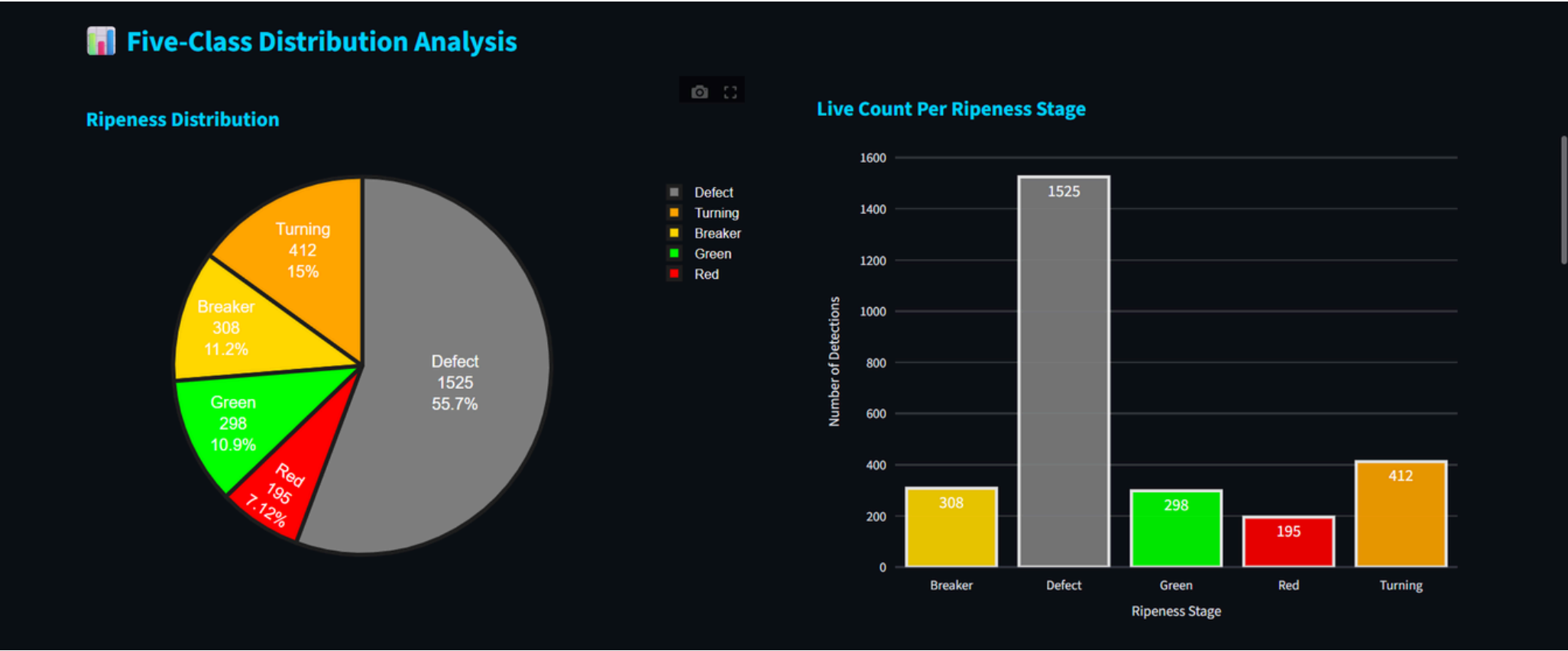
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DEMONSTRATION





INDIVIDUAL CONTRIBUTION



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- Participate in data collection and labeling of the local Padma tomato dataset.
- Lead shelf-life prediction system Development and KPI monitoring integration.
- Support prototype development and testing.
- Contribute to documentation.

INDIVIDUAL CONTRIBUTION



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- Participate in data collection and prototype development.
- Lead model design, tuning, and performance testing.
- Develop the shelf-life prediction component.
- Contribution to documentation and compiling results.

INDIVIDUAL CONTRIBUTION



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- Participate in data collection and prototype development.
- Design and assembled the conveyor and actuator mechanism.
- Integrate sensors and control modules.
- Support system testing and model deployment.
- Contribution to documentation and hardware detailing.



THANK YOU!